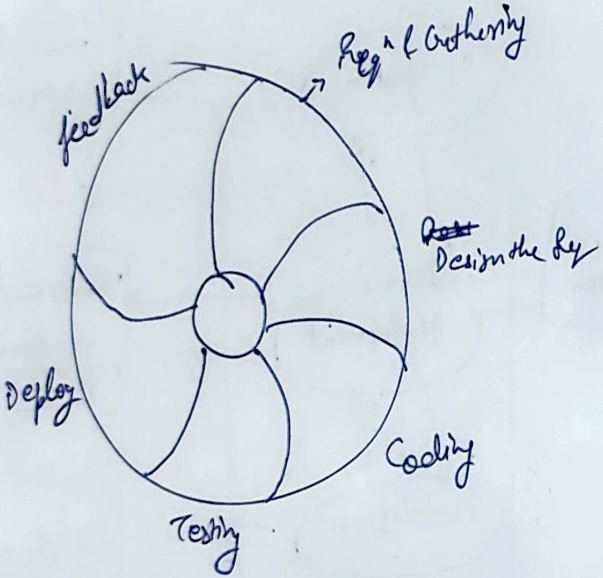
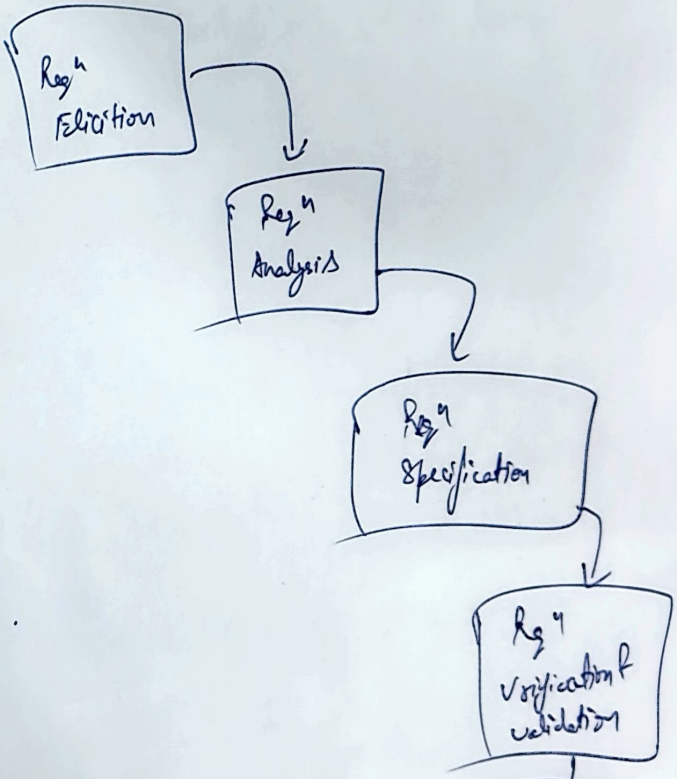
Agile model



Unit-2

Req^n Engg. Process



UNIT-2

Friday
14 Aug, 2026

Requirement Analysis

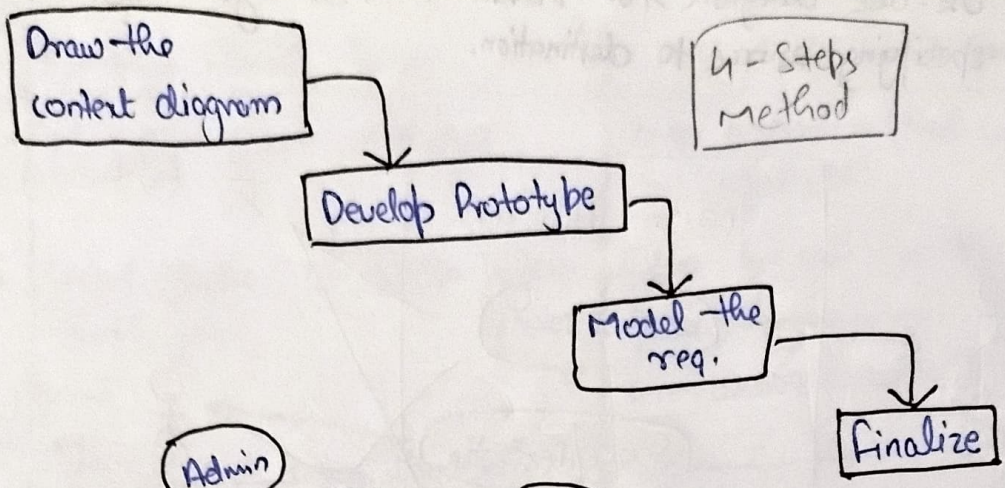
Functional Requirements:
- Specific actions software must be able to perform
- focus on core features

Non-functional Req.:
- Addresses quality & performance
- Speed, quality, scalability, so & so.....

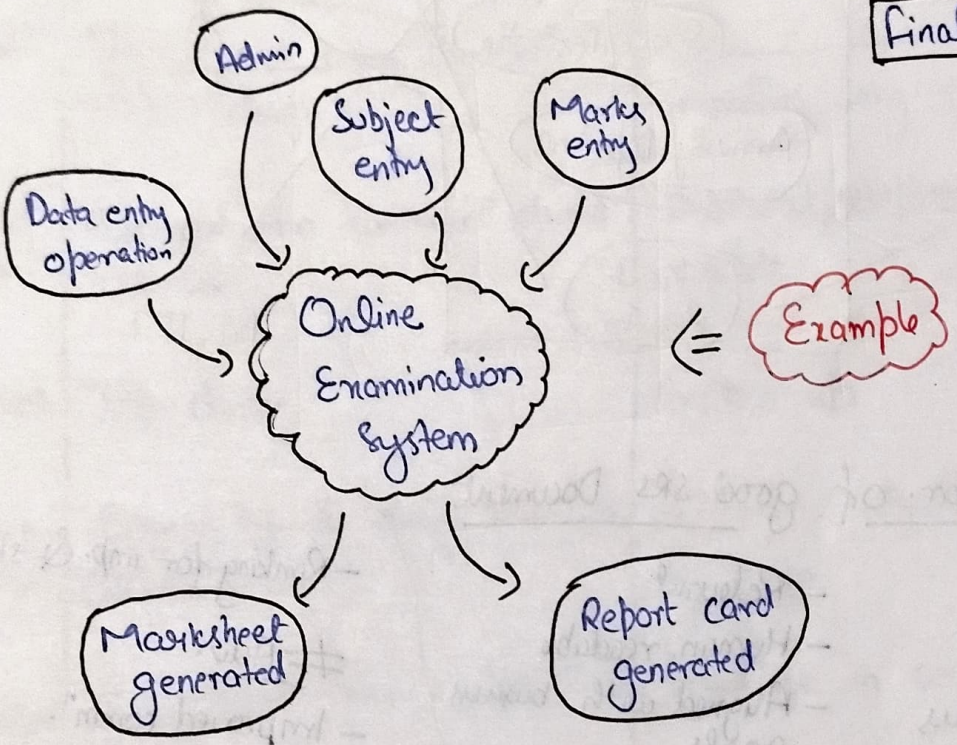
Importance
- Plays critical role
- Foundation
- Clear understanding of stake holders' (customers') needs.
- Avoiding scope creep
- Improved Comm. & colaboration
- Reduced development cost
- Minimized risks & errors
- Better project planning
- Improved quality & user satisfaction.

Steps
1.) Identifying stakeholders & comm. needs
2.) Gathering req.
3.) Analyzing & prioritizing req.
4.) Documenting req.
5.) Validating req.
6.) Creating use cases or user stories
7.) Getting sign-off & approval
8.) Mangging changes.

8-Steps Method



4 - Steps Method

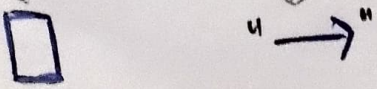


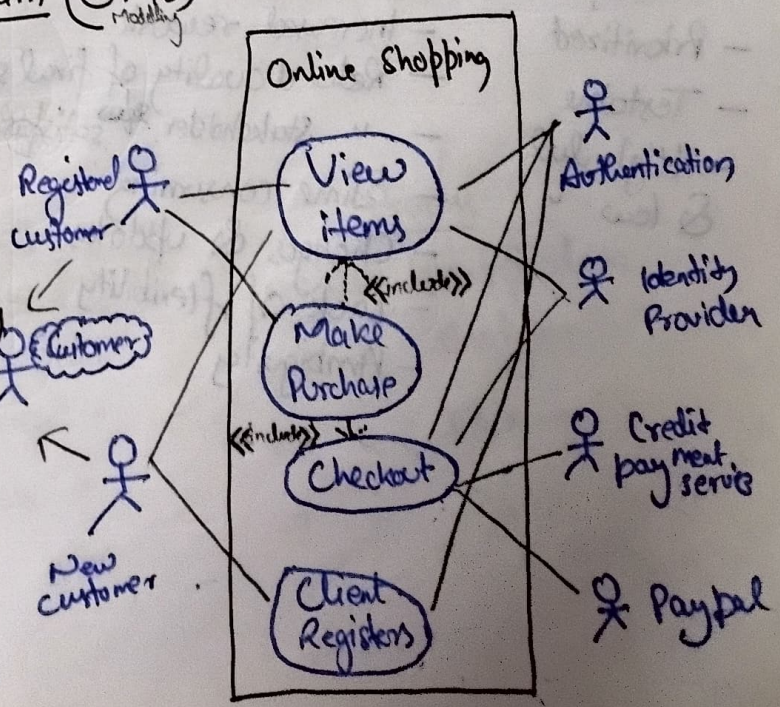
Example

Monday
17 Aug, 2026

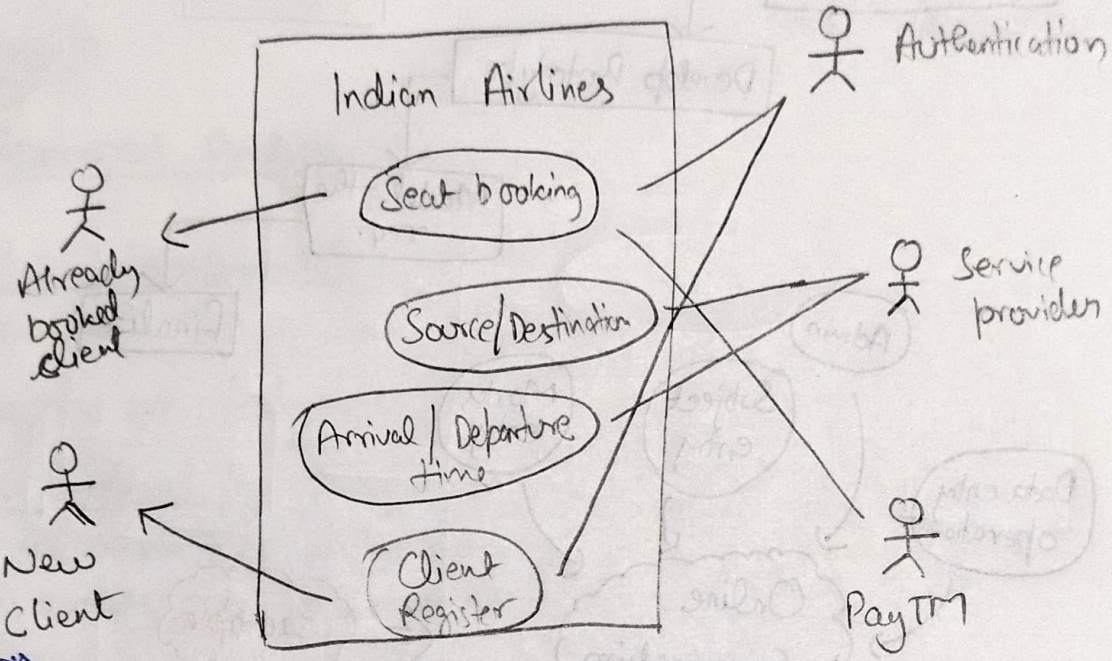
Use-Case Diagram (Unified Modelling Language)

Notation
- Actor
- Use-case → Services provided by system [withdraw, check bal, etc.]
- System-Boundary (Rectangular box outlining inside & outside)
- Association / Relationships (lines connecting actor to use case)





=> Design use-case diagram for Indian Airlines system for a specific flight specifying source to destination.



Wednesday
19 Aug 2026

Quality Char. of good SRS Document

- Complete
- Consistent
- Unambiguous
- Traceable
- Verifiable
- Modifiable
- Prioritized
- Testable
- High lvl & low "
- Relevant
- Human readable
- Aligned with business goals
- Agile methodology

Disadv. -> of poorly written SRS

- Increased rework
- Reduced quality of final s/w
- " stakeholder satisfac.
- Time consuming
- Changes & updates
- Lack of flexibility
- Ambiguity

- Ranking for imp. & stability

Adv.

- Improved comm".
- Increased efficiency
- Improved quality of final s/w system
- Increased stakeholder satisfac.
- Improved traceability & verifie.
- Clarity & completeness
- Reduced rework

Friday
21 Aug, 2026

	Function Oriented Design	Object Oriented Design
Abstration	Main focus → real world func.	Data abstraction (Real world entities are represented)
function	Grped together to obtain higher level func.	Grped together on the basis of data, they operate since classes are associated with their method.
execute	Carried out using structural analysis / design	... unified modeling lang. (UML)
State info	Represented in a centralized shared memory.	Distributed / implemented among objects
Approach	Top-down	Bottom up
Begins basis	Begins by considering use-case diagrams	... identifying objects & classes
Decompose	Decompose in func. or procedural level	Decompose in class level
Use	Computation sensitive application	Evolving system which mimics business or business case.

Monday
24 Aug, 2026

Risk Management

- → uncertainty
- → loss
- → Importance of risk mngmt

Understanding risks in s/w project

- → Project risks
- → Technical "
- → Business "

Process

- → Risk identification
- → " Analysis
- → " Planning
- → " Monitoring